AI Playbook and Toolkit for Public Health Departments

Machine Learning Basics for Epidemiology and Surveillance

ANL 210

Machine learning is a set of methods that allow computer systems to identify patterns in data and use those patterns to classify, predict, cluster, or prioritize information. In public health, machine learning may support surveillance triage, outbreak signal review, risk stratification, case classification, anomaly detection, service planning, or quality improvement. The purpose of this module is not to turn every learner into a data scientist. The purpose is to help public health informaticians, epidemiologists, and technologists understand the concepts needed to evaluate whether a machine learning approach is appropriate, valid, equitable, and operationally useful. Machine learning should be understood as one analytic option among many. Traditional epidemiologic methods, rules-based logic, statistical models, and human judgment remain essential in many public health workflows. A machine learning model may be helpful when patterns are complex, high-dimensional, or difficult to express as simple rules. It may be inappropriate when data are sparse, labels are unreliable, the decision is highly consequential, or the public health question requires causal interpretation rather than prediction. This module introduces supervised learning, unsupervised learning, features, labels, training data, validation data, test data, overfitting, generalization, and model performance. Learners will connect these concepts to public health examples and identify the safeguards needed before machine learning outputs are used to support public health action.

Level	applied/foundational
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling

Learning Objectives

- Explain the difference between supervised, unsupervised, and semi-supervised learning in public health terms.
- Describe features, labels, training data, validation data, and test data.
- Identify when a machine learning approach may be appropriate for a public health workflow and when a simpler method may be preferable.
- Recognize common machine learning risks, including overfitting, biased labels, data leakage, and poor generalization.
- Interpret basic model performance questions in relation to epidemiologic and operational decision-making.
- Produce a machine learning use case review worksheet for a real or realistic public health workflow.

Module Overview

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Why This Topic Matters for Public Health AI

Machine learning matters for public health AI because many proposed tools rely on classification, prediction, ranking, or pattern detection. A surveillance team may want to classify incoming reports by urgency. An epidemiology team may want to detect unusual disease patterns. A program may want to prioritize outreach based on predicted risk. Each of these uses can affect who receives attention, services, investigation, or resources.

Public health staff need enough machine learning literacy to ask whether the model is aligned with the public health purpose. A model built to predict an outcome is not the same as a model that explains why the outcome occurred. A model that performs well in one jurisdiction, population, or reporting environment may not perform well elsewhere. A model trained on historical public health actions may learn historical practice patterns, including under-detection or unequal service access.

The most important public health question is whether the model improves decision-making safely and equitably. This requires more than a performance score. Staff must understand the source data, label quality, model limitations, subgroup performance, workflow impact, human review process, and monitoring plan. Machine learning should strengthen public health practice, not obscure uncertainty or shift responsibility away from accountable staff.

Public Health Example

A health department may consider a machine learning model to classify environmental health complaints by likely urgency. Historical complaint text, location, facility type, prior violations, and resolution outcomes may be used as features. The label might be whether the complaint was ultimately assigned high priority by trained staff. This sounds useful, but the label may reflect staffing capacity or historical prioritization patterns rather than true urgency.

To use this model responsibly, staff would need to evaluate the training data, define the intended use, test performance on recent complaints, review errors by complaint type and community, and decide whether the output is advisory or operational. The model might help sort a queue, but it should not prevent staff from reviewing complaints that do not match historical patterns. The public health example shows that model design decisions become program decisions.

Technical Deep Dive

Machine learning development typically begins with a defined prediction or classification task. The task should be specific enough to test. “Use AI to improve surveillance” is too vague. “Classify incoming reports into routine review, expedited review, and supervisor review based on defined criteria” is more actionable. A clear task allows staff to define the target label, required features, acceptance criteria, and human review process.

Data preparation is often the most important and time-consuming step. Public health data may contain missing values, duplicate records, delayed reporting, inconsistent coding, local terminology, free-text notes, and changes over time. Feature engineering translates raw data into model inputs, but it can also embed assumptions. For example, using prior healthcare utilization as a feature may reflect access to care as much as underlying risk.

Training, validation, and test datasets serve different purposes. Training data are used to fit the model. Validation data are used to tune choices during development. Test data are used to estimate performance on data not used to build the model. Public health teams should be cautious when a vendor reports only development performance without external validation or realistic operational testing.

Model performance should be interpreted in relation to the workflow. A model with high overall accuracy may still miss rare but important events. A model with many false positives may burden staff and create alert fatigue. A model with many false negatives may miss cases, outbreaks, or communities needing support. Thresholds should be selected based on public health consequences, not only technical optimization.

Human review should be designed into the workflow before deployment. Reviewers need to understand what the model output means, what it does not mean, and when to override or escalate. Staff also need feedback channels so model errors can be documented, analyzed, and used to improve monitoring or retraining decisions.

Risks, Failure Modes, and Guardrails

Biased or inconsistent labels. If historical labels reflect unequal detection, inconsistent review, or limited resources, the model may reproduce those patterns. Guardrails include label audits, expert review, subgroup analysis, and documentation of label limitations.

Data leakage. Data leakage occurs when information that would not be available at the time of decision is used during training. Guardrails include temporal validation, careful feature review, and independent testing.

Overfitting and poor generalization. A model may perform well on development data but poorly in a new jurisdiction, time period, or population. Guardrails include held-out testing, external validation, drift monitoring, and conservative deployment.

Misuse of prediction as explanation. A model can predict risk without explaining cause. Guardrails include clear communication, training, and restrictions against causal claims unless supported by appropriate study design.

Application to AI-Supported Workflows

Machine learning may support generative AI, predictive analytics, NLP, anomaly detection, and agentic workflows. In generative AI systems, machine learning concepts help staff understand why outputs may be fluent but unreliable. In predictive workflows, machine learning literacy helps staff evaluate risk scores, thresholds, and subgroup performance. In agentic workflows, models may trigger actions, making validation, permissions, and human approval especially important.

The practical application is disciplined use. Staff should define intended use, validate performance, document limitations, monitor drift, and ensure that human reviewers remain accountable for consequential public health actions. Machine learning should be treated as an analytic support method within a governed workflow.

Reflection Questions

What public health task would the model support: classification, prediction, ranking, clustering, or exploration?

What label would be used, and how reliable or equitable is that label?

Which features might reflect access, documentation, or reporting bias rather than true risk?

What errors would be most harmful in this workflow?

Who would review model outputs, override them, and document concerns?

Practical Exercise

Create a machine learning use case review worksheet for one public health workflow. The worksheet should describe the public health task, intended use, data sources, candidate features, proposed label, likely users, performance measures, subgroup review needs, human review points, and monitoring requirements.

This exercise should produce a governance-ready artifact. Learners should identify at least three reasons the model might fail and at least three safeguards that should be in place before pilot testing.

Expected Artifact or Evidence

Machine learning use case review worksheet

Feature and label inventory

Initial model risk and safeguard list

Human review and monitoring notes

Expected Assignment

- Machine learning use case review worksheet
- Feature and label inventory
- Initial model risk and safeguard list
- Human review and monitoring notes

Knowledge Check

- 1. Which statement best describes supervised learning?
- 2. Why can historical labels create risk?
- 3. What is overfitting?
- 4. What should determine model thresholds in public health workflows?

- 5. What is strong completion evidence for this module?

References and Resources

- CDC Evaluation Framework: https://www.cdc.gov/evaluation/
- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST AI RMF Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html